Sols

1. A spherical snowball melts in such a way that the instant at which its radius is 20 cm, its radius is decreasing at 3 cm/min. At what rate is the volume of the ball of snow changing at that instant?

   (a) Diagram: Draw a picture of the melting snowball. Label the variables of interest.

   (b) Rates: What is the known rate of change? What is the needed rate of change? Include units.

$$\frac{dr}{dt} = -3 \text{ cm/min} \qquad \frac{dV}{dt} = \text{cm}^3/\text{min}$$

   (c) Equation: The rates in the previous part involved the variables V and r. Write an equation from geometry relating V and r.

$$V = \frac{4}{3}\pi r^3$$

   (d) Differentiate: because the snowball is melting, both the radius and volume are really functions of time. Differentiate your formula from the last part with respect to time, t, in minutes.

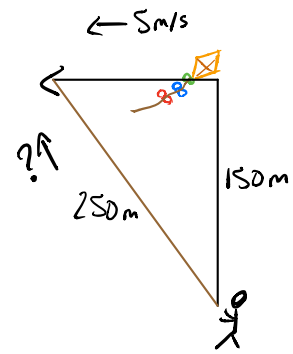
$$\frac{dV}{dt} = 4\pi r^2 \underbrace{\frac{dr}{dt}}_{\text{Chain Rule}}$$

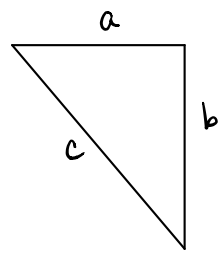
   (e) Substitute and solve: Plug all known quantities into your equation from the last part and solve for the desired rate. Answer the question asked.

$$\frac{dV}{dt} = 4\pi(20)^2(-3) = -4{,}800\pi \ \text{cm}^3/\text{min}$$

2. Omar flies his kite 150m high, where the wind causes it to move horizontally away from him at the rate of 5m per second. In order to maintain the kite at a height of 150m, Omar must allow more string to be let out. At what rate is the string being let out when the length of the string already out is 250m?

   (a) Diagram:





   (b) Rates:

$$\frac{da}{dt} = 5\,m/s \qquad \frac{dc}{dt} = ? \qquad \frac{db}{dt} = 0$$, the height shouldn't change

   (c) Equation:

$$a^2 + b^2 = c^2$$

   (d) Differentiate:

$$2a\frac{da}{dt} + 2b\frac{db}{dt} = 2c\frac{dc}{dt}$$

   (e) Substitute:

$$2a(5) + 2(150)(0) = 2(250)\frac{dc}{dt}$$

$$10a = 500\frac{dc}{dt}$$ Need a.

$$a^2 + (150)^2 = (250)^2$$

$$a = 200$$

   (f) Solve:

$$10(200) = 500\frac{dc}{dt}$$

$$\frac{dc}{dt} = 4\,m/s$$